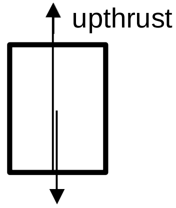
	from point A to point B | 2520
4(a)(i)	At equilibrium, Let the depth of bottom of tube from liquid surface be *h*. Total weight of tube and sand $=$ upthrust $Mg = \rho Ahg$  When tube is displaced downwards and released, Take downwards as positive, resultant force $F = Mg - \rho gA(h + x)$ $= -\rho gAx$
4(a)(ii)	Take downwards as positive, downward displacement is positive, resultant force acting on the tube is upwards. resultant force $F = -\rho gAx$ $Ma = -\rho gAx$ $a = -\left(\dfrac{\rho Ag}{M}\right)x$ (shown) Hence, showing displacement and acceleration are in opposite directions.
4(b)	From $a = -\omega^2 x$, $\omega^2 = \dfrac{\rho Ag}{M}$ $\omega = \sqrt{\dfrac{\rho Ag}{M}}$ $2\pi f = \sqrt{\dfrac{\rho Ag}{M}}$ $f = \dfrac{1}{2\pi}\sqrt{\dfrac{\rho Ag}{M}}$ $= \dfrac{1}{2\pi}\sqrt{\dfrac{(1.2\times10^{3})(5.3\times10^{-4})(9.81)}{0.130}}$ $= 1.10$ Hz
5(a)	By Hooke's law